

# Algebraic Two-Electron Integrals on the Prolate Spheroidal Lattice\*

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Paper 11 established the prolate spheroidal lattice as the natural geometry for diatomic molecules, solving  $H_2^+$  to machine precision (spectral Laguerre solver) with zero free parameters. The extension to two electrons requires electron-electron repulsion integrals  $\langle \phi_i | 1/r_{12} | \phi_j \rangle$  over the six-dimensional two-particle configuration space. The standard approach—numerical quadrature on a grid in prolate spheroidal coordinates—converges slowly due to the Coulomb singularity at  $r_{12} = 0$  and saturates at  $\sim 80\%$  of the exact  $H_2$  dissociation energy  $D_e$  regardless of basis size.

We replace numerical integration with algebraic evaluation using the Neumann expansion of  $1/r_{12}$  in prolate spheroidal coordinates, which decomposes each six-dimensional integral into a finite sum of products of one-dimensional auxiliary integrals computable by recurrence relations. The resulting two-electron integrals are exact within the Neumann truncation order  $L$ , with no six-dimensional quadrature and no fitted parameters. The one step historically left to numerical integration—the log-singular second-kind Legendre moment  $B_l$ —also has a closed form (Sec. IIID): the only transcendental content is the exponential integral  $E_1$ , the Euler–Mascheroni constant  $\gamma$  and  $\ln$ , so *in the  $\sigma$  sector the reduction is fully quadrature-free*. (The  $|m| \geq 1$  channels keep the same kernel; generalising the same  $A_l$ ,  $B_l$ ,  $X_l$  recurrences to associated Legendre functions (Sec. VII) gives a moment-recurrence engine that is *stable* in the  $\delta$  ( $|m| = 2$ ) sector where the differentiation route diverges, and—with the same closed-form seeds—is likewise fully quadrature-free.)

A systematic convergence study on  $H_2$  at  $R = 1.4011$  bohr demonstrates that the Neumann algebraic  $V_{ee}$  yields 92.2% of the exact  $D_e$  with 27 basis functions, compared to 79.6% for numerical  $V_{ee}$  at the same basis size—a uniform 12–20 percentage point improvement across all basis sizes from 1 to 72 functions. The Neumann result plateaus at 92.4%, confirming that  $V_{ee}$  integration error has been eliminated. **[MEASURED]** The remaining 7.6% gap is a basis-space limit, and we identify which one. The basis used here is  $\varphi$ -independent, so it spans only  $m_1 = m_2 = 0$  configurations; but a  $^1\Sigma_g^+$  state constrains the *total*  $M = m_1 + m_2$  to zero, not each  $m_i$  separately, so the  $\pi^2$ ,  $\delta^2$ , ... configurations—every one of them  $^1\Sigma_g^+$ —are absent by construction. Restoring them in the same basis with the same Neumann kernel gives 99.1% of  $D_e$  at  $|m| \leq 1$  (stability envelope 99.0–99.1%; Sec. VII), a gain of 11.6 mHa that no growth along the  $\sigma$  axis reproduces (near-tripling the  $\sigma$  basis is worth 0.34 mHa). **[MEASURED]** That 99.1% is where the *monomial* radial basis  $\xi^j$  stalls—a Hankel moment problem whose overlap matrix is past double precision, not a structural ceiling. Re-basing the same span onto an orthogonal-polynomial family is an exact change of basis that removes the linear dependence, and the energy then climbs monotonically and variationally to 99.81% of  $D_e$  (0.32 mHa, inside chemical accuracy) at a larger truncation and the variationally optimal exponent (Sec. VII A). **[WITHDRAWN]** This paper previously attributed the gap to the electron-electron cusp requiring non-analytic  $r_{12}^{1/2}$  and  $r_{12} \ln r_{12}$  terms, and read that as motivation for hyperspherical coordinates. Neither holds: no such term appears in the calculation above, nor in the grid-based prolate spheroidal FE-DVR treatment of Tao, McCurdy and Rescigno, an accurate two-electron  $H_2$  calculation in these same coordinates. **[MEASURED]** An explicit- $r_{12}$  (James–Coolidge) extension of the algebraic engine, in which every  $r_{12}$  matrix element reduces to the same  $A K_0 - B K_1$  Neumann kernels as  $V_{ee}$  (no quadrature), reaches 0.053 mHa above the exact value—a sixfold reduction of the CI residual—confirming constructively that ordinary integer powers of  $r_{12}$ , not non-analytic terms, close the gap (Sec. VII B). The same reduction extends without a resolution of identity to a heteronuclear diatomic ( $\text{HeH}^+$ ) and to the many-body matrix elements of a two-center four-electron explicit- $r_{12}$  CI, where the genuinely four-body Coulomb-bridge term is evaluated exactly *inside* a real correlation quantity—placing the classical exactness boundary at a two-center three-body integral (Sec. VII B).

## I. INTRODUCTION

The Geometric Vacuum program identifies and discretizes the *natural geometry* of each quantum system—the coordinate system where the Schrödinger equation

separates—replacing continuous integration with algebraic operations on quantum number labels [1–3]. For atoms, the natural geometry is Fock’s three-sphere  $S^3$  [8], where the Coulomb potential disappears into the conformal metric and the Schrödinger equation becomes a pure Laplace–Beltrami eigenvalue problem. For diatomic molecules, the natural geometry is the prolate spheroid, where the two-center Coulomb problem separates into

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\* Paper 12 in the Geometric Vacuum series

coupled ordinary differential equations [4].

Paper 11 implemented this principle for the one-electron molecule  $\text{H}_2^+$ , reaching machine precision (spectral Laguerre solver) with zero free parameters. The extension to two electrons introduces a qualitatively new challenge: the electron-electron repulsion integral

$$V_{ee}^{ij} = \left\langle \phi_i \left| \frac{1}{r_{12}} \right| \phi_j \right\rangle, \quad (1)$$

which couples the coordinates of both electrons and cannot be evaluated by separation of variables. In the prolate spheroidal Hylleraas-type CI framework, these integrals are six-dimensional (three coordinates per electron). The  $\sigma$ -only basis adopted below reduces the effective dimensionality to four; Sec. VI shows that this is a *restriction* on the configuration space, not a symmetry of the  $^1\Sigma_g^+$  state, and measures what it costs.

The standard approach computes Eq. (1) numerically, using the complete elliptic integral  $K(k)$  to perform the azimuthal average of  $1/r_{12}$  and Gauss–Legendre quadrature for the remaining four dimensions  $(\xi_1, \eta_1, \xi_2, \eta_2)$ . This approach has two fundamental limitations:

1. The Coulomb singularity at  $r_{12} = 0$  (the electron-electron coalescence point) is poorly resolved by polynomial quadrature, requiring increasingly fine grids for convergence.
2. The exponential decay  $e^{-\alpha(\xi_1 + \xi_2)}$  of the basis functions extends to  $\xi \rightarrow \infty$ , requiring truncation of the semi-infinite radial domain.

As we demonstrate in Section V, these limitations cause the numerical  $V_{ee}$  to saturate at  $\sim 80\%$  of the exact  $\text{H}_2$  dissociation energy, regardless of how many basis functions are employed.

This paper replaces numerical quadrature with the Neumann expansion of  $1/r_{12}$  in prolate spheroidal coordinates [9–11], which converts each  $V_{ee}$  matrix element into a finite sum of products of one-dimensional integrals. These auxiliary integrals—angular moments  $C_l(p)$ , exponential moments  $A_n(\alpha)$ , and first- and second-kind Legendre moments  $A_l(p, \alpha)$  and  $B_l(p, \alpha)$ —are computed by recurrence relations and closed forms. The log-singular second-kind moment  $B_l$ , once the sole step left to adaptive quadrature, is now itself closed-form (Sec. III D): a single log-moment against  $Q_0$  whose primitive reduces to  $E_1$ ,  $\gamma$  and  $\ln$ . The six-dimensional repulsion integral is thereby reduced, *in the  $\sigma$  sector*, to a finite sum of algebraic operations on quantum number labels with no numerical integration of any dimension; Sec. VII states what the  $|m| \geq 1$  extension costs.

The structure of this paper is as follows. Section II derives the Neumann expansion and its factorization for  $\sigma$ -state matrix elements. Section III defines the auxiliary integrals and their recurrence relations. Section IV describes the computational pipeline. Section V presents the convergence study comparing numerical and algebraic  $V_{ee}$ . Section VI diagnoses the origin of the remain-

ing 7.6% gap and measures its removal. Section VIII discusses implications for the GeoVac program. Section IX concludes.

## II. THE NEUMANN EXPANSION

### A. Expansion of $1/r_{12}$ in prolate spheroidal coordinates

The Coulomb kernel  $1/|\mathbf{r}_1 - \mathbf{r}_2|$  can be expanded in prolate spheroidal coordinates  $(\xi, \eta, \varphi)$  using the Neumann expansion [9, 12]:

$$\frac{1}{r_{12}} = \frac{2}{R} \sum_{l=0}^{\infty} \sum_{m=0}^l (2 - \delta_{m0}) (-1)^m (2l+1) \left[ \frac{(l-m)!}{(l+m)!} \right]^2 P_l^m(\xi_{<}) Q_l^m(\xi_{>}) \quad (2)$$

where  $\xi_{<} = \min(\xi_1, \xi_2)$ ,  $\xi_{>} = \max(\xi_1, \xi_2)$ ,  $P_l^m$  and  $Q_l^m$  are associated Legendre functions of the first and second kind, and  $R$  is the internuclear distance. **[MEASURED]** Earlier versions of this paper printed Eq. (2) without the  $(-1)^m$  and  $(2l+1)$  factors and with the factorial ratio unsquared. That form does not reproduce  $1/r_{12}$ —checked pointwise, it diverges—whereas the expression above agrees with  $1/|\mathbf{r}_1 - \mathbf{r}_2|$  to  $2 \times 10^{-6}$  by  $l = 10$  at well-separated  $\xi$ . The error went undetected because every calculation in *earlier versions* of this paper used only the  $m = 0$  specialisation Eq. (3), which was derived independently—from the Legendre completeness relation, as stated below—rather than specialised from the printed general form. Two of the three discrepancies do vanish at  $m = 0$ ; the  $(2l+1)$  does not, and Eq. (3) carries it because that independent derivation supplies it. Sec. VII is the first use of the general form, which is why the error surfaced only now.

The basis of Eq. (4) below carries no  $\varphi$  dependence, so for *that basis* the azimuthal integration  $\int_0^{2\pi} d\varphi_1 \int_0^{2\pi} d\varphi_2$  retains only the  $m = 0$  component:

$$\left\langle \frac{1}{r_{12}} \right\rangle_{\varphi} = \frac{8\pi^2}{R} \sum_{l=0}^{\infty} (2l+1) P_l(\xi_{<}) Q_l(\xi_{>}) P_l(\eta_1) P_l(\eta_2), \quad (3)$$

where  $P_l \equiv P_l^0$  and  $Q_l \equiv Q_l^0$  are ordinary Legendre functions. The factor  $8\pi^2 = 2 \times 4\pi^2$  combines the  $2/R$  prefactor of the Neumann expansion (2) with the  $4\pi^2$  from  $\int_0^{2\pi} \int_0^{2\pi} \cos 0 d\varphi_1 d\varphi_2$ . The  $(2l+1)$  factor arises from the Legendre polynomial normalization in the angular completeness relation  $\int_{-1}^1 P_l(\eta) P_{l'}(\eta) d\eta = 2\delta_{ll'}/(2l+1)$ .

Equation (3) is the key identity for the  $\sigma$ -only sector: it separates the two-electron Coulomb kernel into a sum of products of functions of individual coordinates. **[SCOPE]** It is *not* the whole kernel for a  $^1\Sigma_g^+$  state. Retaining only  $m = 0$  is a restriction to  $\sigma$  configurations, not a consequence of  $\Sigma$  symmetry: the  $\Sigma$  label fixes  $M = m_1 + m_2 = 0$ , which  $\pi^2$  and  $\delta^2$  configurations satisfy with  $m_2 = -m_1 \neq 0$ . The terms with  $m \neq 0$

in Eq. (2) are precisely what couples those configurations to the  $\sigma$  sector. Sections VI and VII measure the cost of dropping them. Each term in the sum involves  $P_l(\xi_<)Q_l(\xi_>)$ , which couples the two radial coordinates, and  $P_l(\eta_1)P_l(\eta_2)$ , which factorizes completely.

### B. Matrix element factorization

Consider two basis functions of the Hylleraas type in prolate spheroidal coordinates:

$$\phi_i(\xi_1, \eta_1, \xi_2, \eta_2) = \mathcal{N}_i e^{-\alpha(\xi_1+\xi_2)} \xi_1^{j_i} \xi_2^{k_i} \eta_1^{l_i} \eta_2^{m_i} + (1 \leftrightarrow 2), \quad (4)$$

where the  $(1 \leftrightarrow 2)$  exchange ensures bosonic spatial symmetry for the singlet state, and  $(j_i, k_i, l_i, m_i)$  are non-negative integer quantum numbers. The exponential parameter  $\alpha$  controls the spatial extent; for all basis functions in a given calculation we use a common  $\alpha$  optimized variationally.

The  $V_{ee}$  matrix element between  $\phi_i$  and  $\phi_j$  is

$$V_{ee}^{ij} = \iint \phi_i^* \frac{1}{r_{12}} \phi_j J(\xi_1, \eta_1) J(\xi_2, \eta_2) d\xi_1 d\eta_1 d\xi_2 d\eta_2, \quad (5)$$

where  $J(\xi, \eta) = (R/2)^3(\xi^2 - \eta^2)$  is the prolate spheroidal Jacobian (with the  $2\pi$  azimuthal factors already absorbed into Eq. (3)).

Substituting the Neumann expansion (3) and expanding the product of two basis functions, each unsymmetrized term contributes a product of the form

$$e^{-2\alpha(\xi_1+\xi_2)} \xi_1^{P_1} \xi_2^{P_2} \eta_1^{Q_1} \eta_2^{Q_2}, \quad (6)$$

where  $P_1 = j_i + j_j$ ,  $P_2 = k_i + k_j$ ,  $Q_1 = l_i + l_j$ ,  $Q_2 = m_i + m_j$  are the combined powers from bra and ket.

The Jacobian factor  $(\xi^2 - \eta^2)$  for each electron expands the integral into four sub-terms with shifted powers:

$$\begin{aligned} (+1) : & (P_1+2, P_2+2, Q_1, Q_2), \\ (-1) : & (P_1+2, P_2, Q_1, Q_2+2), \\ (-1) : & (P_1, P_2+2, Q_1+2, Q_2), \\ (+1) : & (P_1, P_2, Q_1+2, Q_2+2). \end{aligned} \quad (7)$$

For each such sub-term with effective powers  $(\tilde{P}_1, \tilde{P}_2, \tilde{Q}_1, \tilde{Q}_2)$ , the matrix element factorizes as

$$\frac{\pi^2 R^5}{8} \sum_{l=0}^L (2l+1) X_l(\tilde{P}_1, \tilde{P}_2, 2\alpha) C_l(\tilde{Q}_1) C_l(\tilde{Q}_2), \quad (8)$$

where  $C_l(q) = \int_{-1}^{+1} \eta^q P_l(\eta) d\eta$  are angular moment integrals, and

$$X_l(p_1, p_2, \alpha) = \int_1^\infty \int_1^\infty \xi_1^{p_1} \xi_2^{p_2} e^{-\alpha(\xi_1+\xi_2)} P_l(\xi_<) Q_l(\xi_>) d\xi_1 d\xi_2 \quad (9)$$

is the two-dimensional radial integral coupling the two electrons through the  $P_l(\xi_<)Q_l(\xi_>)$  kernel.

The prefactor  $\pi^2 R^5/8$  arises from  $(2/R) \times (R/2)^6 \times (2\pi)^2 = \pi^2 R^5/8$ , combining the Neumann prefactor  $2/R$ , the two Jacobians  $(R/2)^3$  each, and the azimuthal integration  $4\pi^2$ .

### C. Selection rules

The angular integrals  $C_l(q)$  obey a powerful selection rule:

$$C_l(q) = 0 \quad \text{if } q < l \text{ or } (q-l) \text{ is odd.} \quad (10)$$

This follows from the orthogonality of Legendre polynomials and parity:  $P_l(\eta)$  has parity  $(-1)^l$ , so  $\eta^q P_l(\eta)$  integrates to zero over  $[-1, 1]$  unless  $q+l$  is even, i.e.,  $(q-l)$  is even. The condition  $q \geq l$  ensures that  $\eta^q$  has enough powers to produce a nonzero overlap with  $P_l$ .

For a basis with maximum  $\eta$ -powers  $q_{\max}$ , the effective Neumann truncation is  $l_{\text{eff}} = \min(L, q_{\max})$ , where  $L$  is the nominal truncation order. We use  $L = 20$  throughout, which is more than sufficient since our basis functions have  $q_{\max} \leq 8$ .

## III. AUXILIARY INTEGRALS AND RECURRENCE RELATIONS

The factorized form (8) reduces each  $V_{ee}$  matrix element to products of three families of one-dimensional integrals:  $C_l(q)$ ,  $A_n(\alpha)$  and its Legendre generalizations, and  $B_l(p, \alpha)$ . The  $C_l$  and  $A_n$  families are computed by pure recurrence relations; the log-singular second-kind moment  $B_l$  has a closed form (Sec. IIID), with adaptive quadrature retained only as an independent numerical check.

### A. Angular moments $C_l(q)$

$$C_l(q) = \int_{-1}^{+1} \eta^q P_l(\eta) d\eta. \quad (11)$$

The base cases are

$$C_0(q) = \begin{cases} 2/(q+1) & q \text{ even,} \\ 0 & q \text{ odd,} \end{cases} \quad C_1(q) = \begin{cases} 0 & q \text{ even,} \\ 2/(q+2) & q \text{ odd.} \end{cases} \quad (12)$$

Higher orders follow from the Legendre recurrence:

$$(l+1) C_{l+1}(q) = (2l+1) C_l(q+1) - l C_{l-1}(q). \quad (13)$$

The recurrence  $C_{l+1}(q)$  requires  $C_l(q+1)$ , so computing  $C_{l_{\max}}(q)$  requires workspace up to  $q + l_{\max}$ .

### B. Exponential moments $A_n(\alpha)$

$$A_n(\alpha) = \int_1^\infty \xi^n e^{-\alpha\xi} d\xi. \quad (14)$$

The base case is  $A_0(\alpha) = e^{-\alpha}/\alpha$ , and the upward recurrence is

$$A_n(\alpha) = \frac{n A_{n-1}(\alpha) + e^{-\alpha}}{\alpha}. \quad (15)$$

For negative indices,  $A_{-1}(\alpha) = E_1(\alpha)$  (the exponential integral) and the downward recurrence

$$A_{-n-1}(\alpha) = \frac{e^{-\alpha} - \alpha A_{-n}(\alpha)}{n} \quad (16)$$

extends to arbitrary negative  $n$ .

### C. First-kind Legendre moments $A_l(p, \alpha)$

$$A_l(p, \alpha) = \int_1^\infty \xi^p e^{-\alpha\xi} P_l(\xi) d\xi. \quad (17)$$

The base cases are  $A_0(p, \alpha) = A_p(\alpha)$  and  $A_1(p, \alpha) = A_{p+1}(\alpha)$ , since  $P_0(\xi) = 1$  and  $P_1(\xi) = \xi$ . The Legendre recurrence gives

$$(l+1) A_{l+1}(p, \alpha) = (2l+1) A_l(p+1, \alpha) - l A_{l-1}(p, \alpha). \quad (18)$$

### D. Second-kind Legendre moments $B_l(p, \alpha)$

$$B_l(p, \alpha) = \int_1^\infty \xi^p e^{-\alpha\xi} Q_l(\xi) d\xi. \quad (19)$$

These integrals involve the Legendre function of the second kind  $Q_l(\xi)$ , which has a logarithmic singularity at  $\xi = 1$ :

$$Q_0(\xi) = \frac{1}{2} \ln \frac{\xi+1}{\xi-1}, \quad Q_l(\xi) \sim -\frac{1}{2} \ln(\xi-1) + O(1) \quad \text{as } \xi \rightarrow 1. \quad (20)$$

The log singularity is integrable (the integrand behaves as  $\ln(\xi-1) e^{-\alpha\xi} \rightarrow \ln(\xi-1) e^{-\alpha}$  near  $\xi = 1$ ), but it defeats polynomial quadrature rules such as Gauss-Laguerre, which assume smooth integrands.

Rather than integrate (19) numerically, we reduce it to closed form. Write  $Q_l = P_l Q_0 - W_{l-1}$ , where  $W_{l-1}(\xi) = \sum_{k=1}^l P_{k-1}(\xi) P_{l-k}(\xi)/k$  is a polynomial and

$Q_0(\xi) = \frac{1}{2} \ln \frac{\xi+1}{\xi-1}$ . The polynomial part contributes ordinary exponential moments  $A_n(\alpha)$ ; the entire transcendental content is carried by one family of *log-moments* against  $Q_0$ ,

$$B_l(p, \alpha) = \sum_n (\xi^p P_l)_n L_n(\alpha) - \sum_n (\xi^p W_{l-1})_n A_n(\alpha), \quad L_n(\alpha) = \quad (21)$$

where  $(\cdot)_n$  denotes the coefficient of  $\xi^n$ . The primitive  $L_n$  is itself closed-form. Splitting  $Q_0 = \frac{1}{2}[\ln(\xi+1) - \ln(\xi-1)]$  and substituting  $u = \xi \mp 1$ ,

$$L_n(\alpha) = \frac{1}{2}(L_n^+ - L_n^-), \quad (22)$$

$$L_n^- = e^{-\alpha} \sum_{j=0}^n \binom{n}{j} \frac{j! (H_j - \gamma - \ln \alpha)}{\alpha^{j+1}}, \quad (23)$$

$$L_0^+ = \frac{e^{-\alpha} \ln 2}{\alpha} + \frac{e^\alpha}{\alpha} E_1(2\alpha), \quad L_n^+ = \frac{e^{-\alpha} \ln 2}{\alpha} + \frac{n}{\alpha} L_{n-1}^+ + \frac{1}{\alpha} \left[ \sum_{j=0}^{n-1} \binom{n-1}{j} \frac{j! (H_j - \gamma - \ln \alpha)}{\alpha^{j+1}} \right]$$

with  $H_j$  the harmonic number,  $\gamma$  the Euler-Mascheroni constant, and  $E_1$  the exponential integral. The whole  $B_l$  table then follows from the  $l = 0, 1$  seeds by the Legendre recurrence

$$(l+1) B_{l+1}(p, \alpha) = (2l+1) B_l(p+1, \alpha) - l B_{l-1}(p, \alpha). \quad (24)$$

Thus the log-singular moment introduces *no* numerical integration: its minimal transcendental content is exactly  $\{E_1(2\alpha), \gamma, \ln\}$ , the exponential-integral signature of the Coulomb /  $Q_0$  projection. Adaptive quadrature [20] with subdivision near  $\xi = 1$  is retained only as an independent numerical check, agreeing with Eqs. (21)–(23) to  $\lesssim 10^{-20}$  (backing: `tests/test_paper12_general_m_neumann.py`).

### E. Two-dimensional radial integral $X_l(p_1, p_2, \alpha)$

The integral (9) couples the two radial coordinates through the  $P_l(\xi_<)Q_l(\xi_>)$  kernel. We split it into two ordered regions:

$$X_l(p_1, p_2, \alpha) = \int_1^\infty \int_1^{\xi_2} \xi_1^{p_1} \xi_2^{p_2} e^{-\alpha(\xi_1+\xi_2)} P_l(\xi_1) Q_l(\xi_2) d\xi_1 d\xi_2 + (1 \leftrightarrow 2). \quad (25)$$

The inner integral  $S_l(p, \alpha, \xi_0) = \int_1^{\xi_0} \xi^p e^{-\alpha\xi} P_l(\xi) d\xi$  is the *incomplete* first-kind Legendre moment. Expanding  $\xi^p P_l(\xi) = \sum_j a_j \xi^j$  as a polynomial and integrating each monomial by parts against  $e^{-\alpha\xi}$ :

$$S_l(p, \alpha, \xi_0) = A_l(p, \alpha) e^{-\alpha\xi_0} \sum_j a_j \sum_{k=0}^j \frac{j!}{(j-k)! \alpha^{k+1}} \xi_0^{j-k} + (\text{boundary terms}) \quad (26)$$

Substituting into the outer integral and combining the  $e^{-\alpha\xi_0}$  factor from  $S_l$  with the  $e^{-\alpha\xi_2}$  factor from the outer

integrand yields  $e^{-2\alpha\xi_2}$  terms, which are evaluated as  $B_l(\cdot, 2\alpha)$ :

$$X_l = A_l(p_1) B_l(p_2) + A_l(p_2) B_l(p_1) - (\text{correction terms involving } B_{ee}(\cdot, 2\alpha)) \quad (27)$$

The full expression involves sums over the polynomial coefficients of  $\xi^{p_i} P_l(\xi)$  and is implemented as described in Section IV.

## IV. IMPLEMENTATION

### A. The algebraic $V_{ee}$ pipeline

The computation of the  $V_{ee}$  matrix proceeds in four stages:

1. **Table precomputation.** For a given exponential parameter  $\alpha$  and basis quantum numbers, compute:

- The angular moment table  $C_l(q)$  for  $l \leq l_{\text{eff}}$ ,  $q \leq 2q_{\text{max}} + 2$ , via the recurrence (13). Cost:  $O(l_{\text{max}} \cdot q_{\text{max}})$ , negligible.
- The first-kind Legendre moments  $A_l(p, 2\alpha)$  via the recurrence (18). Cost:  $O(l_{\text{max}} \cdot p_{\text{max}})$ .
- The second-kind Legendre moments  $B_l(p, 2\alpha)$  and  $B_l(p, 4\alpha)$  in closed form (Eqs. (21)–(24)). Cost:  $O(l_{\text{max}} \cdot p_{\text{max}})$  arithmetic, no quadrature; this replaced the former adaptive-quadrature seeding, which had dominated the  $V_{ee}$  build ( $\sim 40\times$  faster, identical values).

2.  **$X_l$  table.** Compute  $X_l(p_1, p_2, 2\alpha)$  for all  $l \leq l_{\text{eff}}$ ,  $p_1, p_2 \leq p_{\text{max}} + l_{\text{max}} + 2$ , using Eq. (27) with the precomputed  $A_l$  and  $B_l$  tables. Cost:  $O(l_{\text{max}} \cdot p_{\text{max}}^2 \cdot l_{\text{max}})$  arithmetic operations.

3. **Matrix assembly.** For each pair  $(i, j)$  of basis functions, expand the bra-ket product into unsymmetrized terms, apply the Jacobian expansion (7), and sum the Neumann series (8) using precomputed  $C_l$  and  $X_l$  tables. Cost per element:  $O(l_{\text{max}})$  (a single loop over  $l$  with table lookups).

4. **Symmetrization.** Exploit  $V_{ee}^{ij} = V_{ee}^{ji}$  to compute only the upper triangle.

The total cost scales as  $O(N_{\text{bf}}^2 \cdot l_{\text{max}})$  for the matrix assembly, plus  $O(l_{\text{max}}^2 \cdot p_{\text{max}}^2)$  for table precomputation. For  $N_{\text{bf}} = 27$  and  $l_{\text{max}} = 8$  (the effective truncation), the  $V_{ee}$  matrix is computed in  $\sim 1$  s.

### B. Hybrid Hamiltonian: Numba $T + V_{ne}$ plus algebraic $V_{ee}$

The one-electron integrals (kinetic energy  $T$  and nuclear attraction  $V_{ne}$ ) are computed analytically using integration by parts for the kinetic energy derivatives

and direct quadrature for  $V_{ne}$ , accelerated by Numba JIT compilation [19]. A dedicated Numba kernel `hamiltonian_tvne_p0_kernel` computes  $T + V_{ne}$  without  $V_{ee}$ , avoiding the expensive elliptic integral computation that dominates the full numerical kernel.

The complete Hamiltonian is assembled as

$$H = H_{T+V_{ne}}^{(\text{Numba})} + V_{ee}^{(\text{Neumann})}, \quad (28)$$

where the first term uses grid-based quadrature (for the non-singular  $T$  and  $V_{ne}$  integrals) and the second uses the algebraic Neumann expansion (for the singular  $V_{ee}$  integrals). This hybrid approach plays to the strengths of each method: grids handle smooth integrands efficiently, while the Neumann expansion handles the Coulomb singularity exactly.

### C. Computational cost comparison

For the  $V_{ee}$  matrix alone:

- **Numerical** (elliptic integral quadrature):  $O(N_{\text{bf}}^2 \cdot N_{\xi}^2 \cdot N_{\eta}^2)$ . At  $N_{\xi} = 30$ ,  $N_{\eta} = 20$ :  $\sim 3$  s for  $N_{\text{bf}} = 6$ , scaling as  $N_{\xi}^2 N_{\eta}^2$  per element.
- **Algebraic** (Neumann expansion):  $O(N_{\text{bf}}^2 \cdot l_{\text{max}} + l_{\text{max}}^2 \cdot p_{\text{max}}^2)$ , pure arithmetic throughout—the closed-form  $B_l$  tables (Sec. III D) removed the former quadrature entirely.

The algebraic approach is both faster (the closed-form  $B_l$  made the  $V_{ee}$  build  $\sim 40\times$  faster than the earlier quadrature-seeded path) and *exact within the Neumann truncation*, whereas the numerical approach always carries grid truncation error.

## V. CONVERGENCE STUDY

### A. Computational setup

We compute the  $\text{H}_2$  ground-state energy at  $R = 1.4011$  bohr (near-equilibrium) using a Hylleraas-type CI with basis functions (4). The basis is parameterized by  $(j_{\text{max}}, l_{\text{max}})$ , which control the maximum powers of  $\xi$  and  $\eta$  respectively. The exponential parameter  $\alpha$  is optimized variationally at each basis size using golden-section search over  $\alpha \in [0.5, 2.5]$ .

For each basis size, we compute the ground-state energy using both numerical and Neumann  $V_{ee}$ :

- **Numerical:** Full grid quadrature with elliptic integral  $K(k)$  on a  $30 \times 20$  grid ( $N_{\xi} \times N_{\eta}$ ).
- **Neumann:** Algebraic  $V_{ee}$  via Eq. (8) with  $L = 20$ .

The exact non-relativistic  $\text{H}_2$  energy at this geometry is  $E_{\text{exact}} = -1.174475$  Ha [14], giving a dissociation energy  $D_e = 0.174475$  Ha relative to two separated hydrogen atoms ( $E = -1.0$  Ha).

## B. Results

Table I presents the complete convergence study. The basis configurations range from a single function ( $j = 0$ ,  $l = 0$ ,  $N = 1$ ) to 72 functions ( $j = 3$ ,  $l = 3$ ).

Three features of Table I deserve emphasis:

*a. 1. Systematic improvement.* The Neumann  $V_{ee}$  yields a lower (better) energy than numerical  $V_{ee}$  at *every* basis size, by 12–20 percentage points. This is not a sporadic improvement from fortuitous cancellation; it is a uniform advantage arising from the elimination of grid truncation error in  $V_{ee}$ .

*b. 2. Numerical plateau at  $\sim 80\%$ .* The numerical  $V_{ee}$  saturates at  $\sim 80\%$   $D_e$  from  $N = 27$  onward: adding basis functions from 27 to 72 improves the energy by only 0.5 percentage points ( $79.6\% \rightarrow 80.1\%$ ). This saturation is caused by grid truncation error in the  $V_{ee}$  integrals, which introduces a systematic positive bias in the electron-electron repulsion. The basis is flexible enough to represent a better wavefunction, but the inaccurate  $V_{ee}$  prevents it from being realized variationally.

*c. 3. Neumann plateau at 92.4%.* The Neumann  $V_{ee}$  also plateaus, but at a much higher level: 92.2% at  $N = 27$  and 92.4% at  $N = 72$ . Since the Neumann  $V_{ee}$  is exact (within the  $L = 20$  truncation, which is converged by the selection rule), this plateau is *not* caused by integration error. It identifies a genuine basis completeness limit, analyzed in Section VI.

## C. The sweet spot: 27 basis functions

The data reveal a clear efficiency optimum at  $N = 27$  ( $j = 2$ ,  $l = 2$ ): this configuration achieves 92.2%  $D_e$  in 705 s, whereas tripling the basis to  $N = 72$  gains only 0.2 percentage points at  $7.5\times$  the computational cost. The diminishing returns beyond  $N = 27$  reflect the fact that additional  $\sigma$ -type basis functions with higher  $\xi$  and  $\eta$  powers cannot access the missing physics—which is not in the  $\sigma$  sector at all (Sections VI and VII).

## D. Grid dependence of numerical $V_{ee}$

To confirm that the numerical plateau is grid-limited, we performed a grid convergence study for the 6-function basis ( $j = 1$ ,  $l = 0$ ) at fixed  $\alpha = 1.0$ :

With the same 6 basis functions,  $D_e\%$  increases from 66% to 85.7% as the grid is refined from  $12 \times 8$  to  $60 \times 40$ . The convergence is slow ( $\sim 1.3$  percentage points per grid doubling) and the cost grows quadratically. In contrast, the Neumann result for this basis is grid-independent at 61.8%  $D_e$  (the basis itself limits accuracy at 6 functions; the  $V_{ee}$  angular series terminates exactly, and the  $B_l$  moment is closed-form, so no quadrature enters).

## VI. DIAGNOSIS OF THE REMAINING 7.6% GAP

### A. What the plateau tells us

The Neumann plateau at 92.4% establishes that the  $V_{ee}$  integration is converged: the Neumann series at  $L = 20$  is exact for our basis (the selection rule (10) caps the effective  $l$  well below 20). The 7.6% gap is therefore a property of the *basis itself*, not of the integral evaluation.

The basis functions (4) are products of integer powers of  $\xi$  and  $\eta$  times a common exponential  $e^{-\alpha\xi}$ . These are smooth, analytic functions of the single-electron coordinates, and—the point that matters here—they carry no  $\varphi$  dependence at all.

**[MEASURED]** The convergence study itself shows that the gap is not reachable along the directions the basis does span. Going from  $N = 27$  to  $N = 46$  to  $N = 72$  moves the energy  $92.2\% \rightarrow 92.4\% \rightarrow 92.4\%$ : 0.34 mHa for  $2.7\times$  the functions. Whatever is missing, more  $\sigma$  functions do not supply it. Sections VIB and VII identify it and remove it.

### B. The electron-electron cusp is real, but it is not this gap

The exact two-electron wavefunction  $\Psi(\mathbf{r}_1, \mathbf{r}_2)$  satisfies the Kato cusp condition [16] at the electron-electron coalescence point  $r_{12} = 0$ :

$$\left. \frac{\partial \bar{\Psi}}{\partial r_{12}} \right|_{r_{12}=0} = \frac{1}{2} \bar{\Psi}(r_{12} = 0), \quad (29)$$

where  $\bar{\Psi}$  is the spherically averaged wavefunction. This cusp produces a discontinuity in the first derivative of  $\Psi$  with respect to  $r_{12}$  and, more fundamentally, introduces non-analytic dependence on  $r_{12}$ .

The Fock expansion [17, 18] of the exact wavefunction near the two-electron coalescence point has the form

$$\Psi = \sum_{n=0}^{\infty} \sum_{k=0}^{\lfloor n/2 \rfloor} R^{n/2} (\ln R)^k f_{nk}(\hat{\Omega}), \quad (30)$$

where  $R = (r_1^2 + r_2^2)^{1/2}$  is the hyperradius and  $\hat{\Omega}$  represents the hyperangular coordinates. The  $R^{1/2}$  and  $R \ln R$  terms are fundamentally non-analytic: no finite sum of polynomial  $\times$  exponential functions in prolate spheroidal coordinates can represent them *exactly*.

**[WITHDRAWN]** Earlier versions of this paper concluded from this that the cusp is what caps the calculation at 92.4%. It is not, and the argument was never sound: non-analyticity implies slow convergence of a partial-wave expansion, not a floor at 7.6%. Two calculations settle it. Tao, McCurdy and Rescigno [7], working in these same prolate spheroidal coordinates with a polynomial angular basis and no  $r_{12}$  dependence anywhere in

TABLE I. Convergence of  $\text{H}_2$  ground-state energy at  $R = 1.4011$  bohr.  $N$  is the number of basis functions.  $D_e\%$  is the fraction of the exact dissociation energy  $D_e = 0.174475$  Ha recovered.  $\Delta E = E_{\text{Neumann}} - E_{\text{numerical}}$ .

| Config     | $N$ | $E_{\text{num}}$ (Ha) | $D_e\%_{\text{num}}$ | $t_{\text{num}}$ (s) | $E_{\text{Neu}}$ (Ha) | $D_e\%_{\text{Neu}}$ | $t_{\text{Neu}}$ (s) | $\Delta E$ (Ha) |
|------------|-----|-----------------------|----------------------|----------------------|-----------------------|----------------------|----------------------|-----------------|
| $j=0, l=0$ | 1   | -1.073192             | 41.9                 | 1                    | -1.107907             | 61.8                 | 3                    | -0.0347         |
| $j=1, l=0$ | 3   | -1.083438             | 47.8                 | 8                    | -1.115282             | 66.1                 | 9                    | -0.0318         |
| $j=1, l=1$ | 6   | -1.103485             | 59.3                 | 32                   | -1.129606             | 74.3                 | 32                   | -0.0261         |
| $j=2, l=1$ | 12  | -1.107960             | 61.9                 | 152                  | -1.132373             | 75.9                 | 131                  | -0.0244         |
| $j=2, l=2$ | 27  | -1.138836             | 79.6                 | 767                  | -1.160961             | 92.2                 | 705                  | -0.0221         |
| $j=3, l=2$ | 46  | -1.139329             | 79.8                 | 2248                 | -1.161162             | 92.4                 | 1907                 | -0.0218         |
| $j=3, l=3$ | 72  | -1.139792             | 80.1                 | 6076                 | -1.161304             | 92.4                 | 5271                 | -0.0215         |

TABLE II. Grid convergence of numerical  $V_{ee}$  for the 6-function basis at  $R = 1.4011$  bohr,  $\alpha = 1.0$ .

| Grid ( $N_\xi \times N_\eta \times N_\phi$ ) | $D_e\%$ | $t$ (s) |
|----------------------------------------------|---------|---------|
| $12 \times 8 \times 8$                       | 66.0    | 1       |
| $20 \times 14 \times 16$                     | 76.2    | 6       |
| $30 \times 20 \times 24$                     | 81.4    | 45      |
| $40 \times 26 \times 16$                     | 83.7    | 83      |
| $60 \times 40 \times 16$                     | 85.7    | 447     |

it, carry out an accurate two-electron  $\text{H}_2$  calculation in these same coordinates. (Their odd- $m$  radial functions do carry an explicit  $(\xi^2 - 1)^{1/2}$ , which they introduce for the non-analytic behaviour at  $\xi = 1$ ; that is a one-electron boundary condition on the focal axis, the analogue of the  $\sin \theta$  behaviour of ordinary spherical harmonics at odd  $m$ , and is unrelated to  $r_{12}$ . The point stands as stated: nothing in their basis represents the *interelectronic* non-analyticity.) And the calculation reported in Sec. VII below, run in the basis of Eq. (4) with the same Neumann kernel and likewise no  $r_{12}$  dependence, reaches 99.1%. Neither could exist if representing the cusp were the binding constraint at this accuracy. What the cusp does cost is the slow tail of the partial-wave series; at  $|m| \leq 1$  and  $l \leq 3$  that residue is under one percent of  $D_e$ .

### C. Why $r_{12}$ basis functions help only partially

The traditional approach to the cusp problem is to include explicit  $r_{12}$  dependence in the basis, as in the classic James-Coolidge [13] and Kolos-Wolniewicz [14] wavefunctions. We tested this by extending the basis with  $r_{12}^p$  ( $p = 1, 2$ ) factors. The results are instructive: with 18 functions including  $r_{12}^2$  and angular terms ( $l \leq 1$ ), the numerical  $V_{ee}$  reaches 86.8%  $D_e$  at the  $20 \times 14 \times 16$  grid—but this *also* saturates with grid refinement. The  $r_{12}$  factors improve the wavefunction’s ability to represent the cusp, but the  $V_{ee}$  integrals involving  $r_{12}$  are even harder to evaluate numerically (they become seven-dimensional integrals with an additional radial singularity).

The Neumann algebraic approach at 92.2% with  $p = 0$  basis functions outperforms the  $r_{12} +$  numerical ap-

proach at 86.8% with  $p \leq 2$  basis functions. What that comparison establishes is narrower than it looks: *an exact  $V_{ee}$  beats  $r_{12}$  basis functions carrying grid error*. It says nothing about the cusp in either direction. (Earlier versions drew the stronger conclusion that “the cusp does not need  $r_{12}^p$  basis functions—it needs exact  $V_{ee}$ ”; that goes with the rest of the withdrawn cusp reading, and Sec. VIB gives the cusp’s actual cost here.) The  $r_{12}$  basis functions are a workaround for poor  $V_{ee}$  integration, not a solution to the underlying representation problem.

### D. The missing channels

[MEASURED] The 7.6% gap is a *configuration-space* limitation, and an elementary one. A  $^1\Sigma_g^+$  state requires the *total* azimuthal quantum number  $M = m_1 + m_2$  to vanish. It does not require  $m_1 = m_2 = 0$ . The configurations  $\pi^2$  ( $m_1 = +1, m_2 = -1$ ),  $\delta^2$  ( $m_1 = +2, m_2 = -2$ ), and so on all have  $M = 0$  and are all  $^1\Sigma_g^+$ ; they carry the *angular* part of the electron correlation, the left-right and in-out parts being what a  $\sigma$ -only space already describes. The basis Eq. (4) is  $\varphi$ -independent and therefore excludes every one of them, and Eq. (3) drops exactly the kernel terms that would couple them in. The two restrictions are consistent with each other, which is why the calculation converges cleanly to a wrong answer.

Section VII restores the channels and measures the result: 92.4%  $\rightarrow$  99.1% of  $D_e$ . What remains after that is ordinary basis incompleteness—higher  $l$ , higher  $n$ , and the  $\delta$  channels. That remainder is  $\approx 1.6$  mHa in total, of which the  $\delta$  channels account for  $\approx 0.5$  mHa on an independent Gaussian-basis estimate; the balance is not itemised here.

For completeness, the interelectron coordinate in prolate spheroidals is a function of all five relative variables:

$$r_{12}^2 = \frac{R^2}{4} [(\xi_1^2 - 1)(1 - \eta_1^2) + (\xi_2^2 - 1)(1 - \eta_2^2) - 2\sqrt{(\xi_1^2 - 1)(1 - \eta_1^2)(\xi_2^2 - 1)(1 - \eta_2^2)}] \quad (31)$$

(The  $\cos(\varphi_1 - \varphi_2)$  factor was omitted in earlier versions of this equation, which therefore held only at  $\varphi_1 = \varphi_2$ .)

Note the  $\cos(\varphi_1 - \varphi_2)$  dependence: the interelectron distance is not a function of  $\xi_1, \eta_1, \xi_2, \eta_2$  alone. A  $\varphi$ -independent basis cannot resolve it, which is the same restriction stated in coordinates.

[**WITHDRAWN**] Earlier versions of this paper closed this section by identifying hyperspherical coordinates  $(\rho, \theta, \hat{\Omega})$  as the natural geometry the gap called for, on the grounds that the coalescence cusp has a simple representation there. That inference rested on the cusp diagnosis withdrawn above and does not survive it. Hyperspherical coordinates remain the right setting for genuine three-body coalescence problems, and Paper 13 [5] uses them for helium on independent grounds; what is withdrawn is the claim that *this* calculation’s residual motivates them.

## VII. RESTORING THE AZIMUTHAL CHANNELS

[**MEASURED**] We extend the basis of Eq. (4) with an azimuthal quantum number,

$$u_{jl\mu}(\xi, \eta) = \xi^j \eta^l (\xi^2 - 1)^{\mu/2} (1 - \eta^2)^{\mu/2} e^{-\alpha\xi}, \quad \Phi_\mu = u u' \cos \mu(\varphi_1 - \varphi_2), \quad (32)$$

pairing  $m_1 = +\mu$  with  $m_2 = -\mu$  through  $\cos \mu(\varphi_1 - \varphi_2)$ —the  $M = 0$ ,  $^1\Sigma_g^+$  combination, which for  $\mu = 1$  is the familiar  $\pi_x\pi_x + \pi_y\pi_y$ . At  $\mu = 0$  the factor  $u$  reduces to the one-electron factor of Eq. (4) and  $\Phi_0$  to its symmetrised product, so the  $\mu = 0$  sector is exactly the original basis. The overlap, kinetic and nuclear-attraction matrices remain *exact*—every integrand is a polynomial in  $\xi$  and  $\eta$  times  $e^{-2\alpha\xi}$ , evaluated against the moments  $A_n$  of Sec. III and elementary  $\eta$  moments—and all three are diagonal in  $\mu$ .  $V_{ee}$  uses the full kernel Eq. (2); its  $m \neq 0$  terms are what couple different  $\mu$ .

The  $\eta$  integrals remain exact polynomial moments, because the kernel’s  $(1 - \eta^2)^{m/2}$  and the basis’s  $(1 - \eta^2)^{\mu/2}$  always combine to an integer power. The sum over  $l$  terminates exactly: integrating by parts  $m$  times (the boundary terms vanish since  $(1 - \eta^2)^s$  has a zero of order  $s \geq m$  at  $\eta = \pm 1$ ) shows the moment is identically zero for  $l > Q + 2s - m$ , with  $Q$  the combined  $\eta$  power and  $s = (\mu_i + \mu_j + m)/2$ . That rule must be imposed explicitly rather than left to cancellation: the Legendre derivative coefficients reach  $\sim 10^{10}$ , and their floating-point residue multiplied by the radial integral produced a  $-3 \times 10^8$  Ha artifact before the cutoff was enforced.

Table III gives the result. At the largest basis the azimuthal channels are worth +11.64 mHa, taking  $92.42\% \rightarrow 99.1\%$  of  $D_e$ . The  $\mu = 0$  column reproduces the Neumann column of Table I to 161, 1.6, 1.0, 6.7 and 58  $\mu$ Ha.

Three controls make the reading unambiguous. *First*, the gain is not a basis-count effect: tripling the basis along the  $\sigma$  axis is worth 0.34 mHa (Sec. VI), while opening the azimuthal axis at fixed  $(j_{\max}, l_{\max})$  is worth 11.6 mHa. *Second*, an independent route agrees. A full CI in a Cartesian Gaussian basis—different functions, different integrals, different code—restricted to  $m = 0$  orbitals converges to 92.34%, within 0.2 mHa of this pa-

TABLE III.  $H_2$  at  $R = 1.4011$  bohr with the azimuthal channels restored. Same basis family, same Neumann kernel,  $\alpha$  scanned, canonical orthogonalization throughout — eigenvectors of  $S$  with eigenvalue below  $10^{-11} \lambda_{\max}$  are discarded, and the surviving dimension is given in parentheses where it differs from  $N$ .  $D_e^{\text{exact}} = 0.174475$  Ha.

| $(j_{\max}, l_{\max})$ | $N_\sigma$ | $D_e\%$ ( $\mu = 0$ ) | $N_{ m  \leq 1}$ | $D_e\%$ ( $ m  \leq 1$ ) |
|------------------------|------------|-----------------------|------------------|--------------------------|
| (1, 1)                 | 6          | 74.19                 | 12               | 81.63                    |
| (2, 1)                 | 12         | 75.87                 | 24               | 83.47                    |
| (2, 2)                 | 27         | 92.25                 | 54               | 98.96                    |
| (3, 2)                 | 46         | 92.37                 | 92 (76)          | 99.00                    |
| (3, 3)                 | 72 (65)    | 92.42                 | 144 (115)        | 99.09                    |

per’s  $\sigma$ -only value in a completely unrelated basis; releasing  $|m| \leq 1$  takes it to 99.10%, against 99.1% here—agreement well inside the stability envelope quoted below, and not to be read as a two-decimal coincidence. *Third*, the general- $m$  kernel of Eq. (2) was checked pointwise against  $1/|\mathbf{r}_1 - \mathbf{r}_2|$ , agreeing to better than  $10^{-6}$  at well-separated  $\xi$  by  $l = 10$ . That figure is a point evaluation, not a uniform bound: the Neumann series converges slowly when the two  $\xi$  are close, and the second-kind recursion used here degrades at larger  $l$ , so nearby geometries and higher truncations are worse. What the check establishes is the *form* of the prefactor—the superseded one errs by nine orders of magnitude at the same point—not an accuracy claim for the kernel.

Backing: `tests/test_paper12_azimuthal_channels.py`, whose guards are fire-tested against the specific wrong answers they exclude (a killed azimuthal coupling, the superseded kernel prefactor, a disabled canonical orthogonalization, and a truncation cap that would hide the selection rule).

[**MEASURED**] Both limits stated in earlier versions of this section have since been removed. The  $\mu > 0$  radial integrals are now available from the  $A_l$ ,  $B_l$ ,  $X_l$  recurrences of Sec. III generalised to the associated Legendre functions (module `geovac/neumann_vee_general_m.py`). The first-kind moment  $A_l^{m,s}(p) = \int_1^\infty \xi^p (\xi^2 - 1)^s e^{-c\xi} [d^m P_l / d\xi^m] d\xi$  is a closed-form polynomial contraction against the  $A_n$ ; the second-kind moment  $B_l^{m,s}$ —whose integrand is singular as  $(\xi - 1)^{-m}$  at  $\xi = 1$ , regularised by the weight  $(\xi^2 - 1)^s$  kept *intact* rather than expanded into  $\xi$ -powers (the bare  $\xi^p d^m Q_l$  moment diverges for  $m \geq 1$ )—is built by the associated-Legendre forward  $l$ -recurrence  $(l - m + 1)B_{l+1}(p) = (2l + 1)B_l(p + 1) - (l + m)B_{l-1}(p)$ , seeded at  $l = m, m + 1$ ; the ordered- $\xi$  integral  $X_l^{m,s}$  then follows from the same integration-by-parts split as the  $\sigma$  case. The engine reduces to `geovac.neumann_vee` at  $m = 0$  to  $\sim 10^{-9}$  elementwise, matches a high-precision reference for  $X_l^{m,s}$  to better than  $10^{-9}$  through  $l = 8$ ,  $m \leq 4$ , and reproduces the grid result at (2, 2),  $|m| \leq 1$  to the last printed digit. Its decisive property is *stability* in the  $\delta$  ( $|m| = 2$ ) sector: the fourth-derivative  $Q_l$  cancellation near  $\xi = 1$  drives the differentiation route to a non-



variational  $\approx -14.5$  Ha, whereas the recurrence build is variational and supplies the  $\delta$  channels from the native prolate route—a +0.25 mHa gain at (2, 2), consistent with the  $\approx 0.5$  mHa Gaussian estimate (the (2, 2)  $\delta$  subspace being the smaller). The engine is *fully quadrature-free*: the low- $l$   $B_l^{m,s}$  seeds are closed-form by the same construction as the  $\sigma$  sector (Sec. III D). The obstruction that had kept these seeds numerical—the individual pole terms  $d^k Q_0 \sim (\xi - 1)^{-k}$  of  $d^m Q_l$  have divergent moments—dissolves because  $s \geq m$  always holds ( $s = (\mu_i + \mu_j + m)/2$  with  $m \in \{\mu_i + \mu_j, |\mu_i - \mu_j|\}$ ): the intact weight  $(\xi^2 - 1)^s = (\xi - 1)^s (\xi + 1)^s$  then *fully polynomialises* every pole ( $k = m - a \leq m \leq s$ ), leaving a single log-moment against  $Q_0$  whose primitive is Eqs. (21)–(23). The moment tables are still built in extended precision (the associated Legendre coefficients cancel over several orders), but no numerical integration remains. Backing: `tests/test_paper12_general_m_neumann.py`.

[**MEASURED**] Finally, a caution about the  $\sigma$  column itself. The overlap matrix of this basis family is strongly linearly dependent at high powers and a single  $\alpha$ : measured  $\text{cond}(S) = 2.6 \times 10^{14}$  at  $(j, l) = (3, 3)$ ,  $\mu = 0$  (the  $N = 72$  basis of Table I), rising to  $2.0 \times 10^{16}$  once  $\mu = 1$  doubles the basis. Beyond  $\sim 10^{16}$  a direct generalised eigensolve is not trustworthy—at  $N = 144$  it returned  $-79$  Ha—so Table III uses canonical orthogonalization throughout.

[**MEASURED**] Canonical orthogonalization *bounds* that pathology; it does not remove it, and we state the envelope rather than leave the reader to assume a converged four-figure number. At the  $|m| \leq 1$ ,  $N = 144$  basis the solver still returns non-variational values at a *majority* of  $\alpha$ . Measured at the stated threshold over  $\alpha \in [0.90, 1.30]$ , six of nine grid points fail. The failures are not marginal:  $\alpha = 1.15$  and  $1.20$  return  $-4.1$  and  $-8.1$  Ha,  $\alpha = 0.95$  returns  $-277$  Ha, and  $\alpha = 1.00$ —the natural default, and the value a reader is most likely to try first—returns  $-64.3$  Ha. The three that survive give 99.03, 99.09 and 99.05% at  $\alpha = 1.10, 1.25, 1.30$ . So the reported value is the best *variational* point of an  $\alpha$  scan, and points failing the variational test are discarded—a selection the reader should see, not infer. The value also moves with the discard threshold: at  $\alpha = 1.25$  it reads 99.14% at  $10^{-12}$ , 99.09% at  $10^{-11}$ , 98.99% at  $10^{-10}$  and 98.41% at  $10^{-8}$ , and is non-variational below  $\sim 10^{-12}$ . The honest statement is therefore 99.0–99.1%, with 99.09% the best variational value at the stated parameters.

[**SCOPE 2026-09-14, corrected twice**] Two earlier versions of this passage overstated how stable that number is. The first called the conclusion “insensitive to the choice” [retracted 2026-09-14: p12-envelope-insensitive] and then quoted the figure that refutes it. The second replaced it with a range, “95.5–99.1% across the grid”, and called the qualitative conclusion “robust across the grid” [retracted 2026-09-14: p12-grid-floor-955]. Both are wrong, and the second is wrong in the direction that flatters the result.

95.5% is the cell at  $\alpha = 1.10$ , threshold  $10^{-8}$ . It

is the weakest variational value only if  $\alpha$  is restricted to  $\geq 1.10$ —which excludes precisely the low- $\alpha$  half this paragraph has just identified as pathological. Over the full declared range  $\alpha \in [0.90, 1.30]$  the weakest variational value is 76.11%, at  $\alpha = 0.90$  and threshold  $10^{-8}$ ; and at that same cell the  $\sigma$ -only solve returns 91.06%, so opening the azimuthal channels there makes the answer *worse* by fifteen points. The qualitative conclusion is therefore *not* robust pointwise across the grid, and this paper does not claim that it is.

What the result rests on instead, stated exactly: at the tight discard thresholds, where the generalized eigenproblem is actually being solved rather than regularized, every surviving variational point lies between 99.03 and 99.14%; and the independent Gaussian-basis route, which is well conditioned and needs no discard threshold at all, gives 99.10%. The claim is the agreement of those two. It is not a property of the whole ( $\alpha$ , threshold) grid, most of which is conditioning noise. The  $-1.161304$  Ha quoted in Table I sits  $58 \mu\text{Ha}$  below the conditioned value, i.e. the last two of its six decimals reflect linear dependence rather than physics. No conclusion in this paper turns on them. That the number is conditioning-*limited* rather than physics-limited is shown directly in Sec. VII A: re-basing this same span removes the linear dependence and the energy climbs to chemical accuracy.

#### A. The monomial cap is conditioning, not a ceiling

[**MEASURED**] The 99.1% above is the ceiling of the *monomial* radial factor  $\xi^j$ , not of the prolate geometry or the electron-electron cusp. The set  $\{\xi^j e^{-\alpha\xi}\}$  is a Hankel moment problem, exponentially ill-conditioned in the degree: the overlap reaches  $\text{cond} = 1.8 \times 10^{16}$  already at (3, 3),  $|m| \leq 1$  and  $4.1 \times 10^{26}$  at (5, 5), so the basis cannot be pushed further in double precision. But the monomials  $\{\xi^j\}_{j \leq J}$  and, e.g., the Laguerre polynomials  $\{L_j(2\alpha(\xi - 1))\}_{j \leq J}$  span the *same* degree- $J$  space, so re-basing onto an orthogonal-polynomial family is an *exact* change of basis: it leaves the span—and hence the exact-arithmetic energy—unchanged, and only the conditioning changes.

[**MEASURED**] Because the congruence  $CSC^\top$  amplifies the  $10^{-16}$  entry error by  $\|C\|^2$  and the high-degree monomial matrices carry  $\text{cond} \sim 10^{26}$ , a float64 change of basis is corrupted—it breaks for *both* families at every truncation tested. Every matrix entry that passes through the monomial basis is therefore built in extended precision, the well-conditioned orthogonal matrices are downcast to float64, and the generalized eigenproblem is solved on the unit-normalized (correlation) matrices—a diagonal congruence that leaves the eigenvalues unchanged but removes the norm-spread inflation of  $\text{cond}$ .

[**MEASURED 2026-09-18**] *Two condition numbers appear in this section and they answer different questions; conflating them is a live trap.* The raw  $\text{cond}$  figures above ( $1.8 \times 10^{16}$ ,  $4.1 \times 10^{26}$ ) are the right metric

for the  $\|C\|^2$  congruence-corruption argument, which is about entry error amplified by the change of basis. They are *not* the metric the eigensolve sees: that is the unit-normalized cond, and the two can differ by six orders on the same matrix. Measured at  $(3,3)$ ,  $|m| \leq 1$ : raw  $3.0 \times 10^{15}$  against normalized  $1.1 \times 10^{11}$ . The practical consequence is not academic—a raw comparison was used to conclude that a two-exponent radial set carried a  $47\times$  conditioning penalty, whereas on the governing metric the same basis is  $544\times$  *better* conditioned than the single-exponent one, because two distinct decay rates make the functions less linearly dependent rather than more. Quote raw cond when arguing about a congruence and normalized cond when arguing about a solve. With that pipeline the energy climbs monotonically and variationally past the monomial wall (Table IV): opening the  $\delta$  channels on the re-based basis reaches 99.71% of  $D_e$  at  $(4,4) + \delta$  and 99.77% (0.41 mHa) at  $(5,5) + \delta$ —inside chemical accuracy, a four-fold reduction of the residual from the monomial 99.09%, every point containing its predecessor and lying below it.

[**MEASURED**] That ladder is computed at a *fixed*  $\alpha = 1.0$ , which is what makes it a ladder: the statement that each point contains its predecessor and lies below it holds only at a consistent exponent, so Table IV keeps  $\alpha$  fixed throughout. But  $\alpha$  is a variational parameter, and 1.0 is not its optimum. Scanning it at the largest truncation gives 0.406, 0.346, 0.32 and 0.334 mHa at  $\alpha = 1.00, 1.20, 1.40$  and  $1.50$ , so the minimum is bracketed at  $\alpha = 1.40$  and the best variational result this basis supports is 99.81% of  $D_e$  (0.32 mHa)—better than the  $\alpha = 1.0$  endpoint by 0.082 mHa, a fifth of that residual. The improvement is not a trade: the optimal point is variational, retains all 1944 functions, and is *more* tightly conditioned than  $\alpha = 1.0$  ( $4.9 \times 10^{10}$  against  $9.4 \times 10^{10}$ ). Both numbers are therefore reported, and they are not competing: 99.77% is the endpoint of the fixed- $\alpha$  ladder, 99.81% is the basis’s best variational point.

[**MEASURED 2026-09-18**] *What the remaining residual is not.* Three knobs of this basis have now been measured against the  $|m| \leq 1$  ladder, one axis at a time, and none of them reaches it: radial degree +1 buys 34  $\mu$ Ha, angular degree +1 buys 8  $\mu$ Ha, and re-optimizing the single exponent buys 0.056 mHa—together about 9% of that rung’s 1.096 mHa residual. A *two-block* exponent set (low degrees at  $a_1$ , high degrees at  $a_2$ , at equal function count) does not reach it either: +0.005 mHa at  $(4,4)$  and  $-0.009$  mHa at  $(3,3)$ , and the  $(4,4)$  optimum is a *plateau* rather than a point—1.107, 1.108, 1.110 mHa across three quite different  $(a_1, a_2)$  pairs, flat to 3  $\mu$ Ha—which is direct evidence that a second exponent has no work to do at this basis size. That last result also disposes of the natural reading of the drift in  $\alpha_{\text{opt}}$  with basis size ( $\sim 1.20$  at  $(4,4)$ ,  $1.40$  at  $(5,5)$ ): the drift is real but it is not relievable strain. We state this as the *elimination* of three knobs, not as a measurement of the cusp: what is established is that more polynomial degree and more exponents do not close the gap in this family, which is

TABLE IV.  $\text{H}_2$  at  $R = 1.4011$  bohr, re-based to orthogonal polynomials (Laguerre $\times$ Legendre and the  $\mu$ -adapted Gegenbauer give the identical energy). Same span as the monomial basis of Table III, re-conditioned;  $\alpha = 1.0$ , the extended-precision  $V_{ee}$  build and unit-normalized solve of the text.  $D_e^{\text{exact}} = 0.174475$  Ha.

| basis / channels                  | $N$  | $D_e\%$ |
|-----------------------------------|------|---------|
| monomial ceiling ( $ m  \leq 1$ ) | 144  | 99.09   |
| re-based $(3,3)$ , $ m  \leq 1$   | 256  | 99.22   |
| re-based $(4,4) + \delta$         | 975  | 99.71   |
| re-based $(5,5) + \delta$         | 1944 | 99.77   |

weaker than—and should not be quoted as—a demonstration that the remainder *is* the electron-electron cusp.

[**MEASURED**] The extended-precision requirement above constrains the *change of basis*, not the orthogonal matrices themselves, and so does not apply to a matrix that is never built in the monomial basis at all. The one-body operators admit such a construction: every one-electron block factorises as radial  $\times$  angular, so  $S$  and  $H_1$  can be assembled directly in the orthogonal basis from one-dimensional blocks small enough to evaluate exactly, after which the  $O(N^2)$  two-electron assembly is ordinary double precision. Built that way they agree with the extended-precision route re-based to  $6 \times 10^{-16}$  of the matrix scale at  $(5,5) + \delta$  in both families, and every energy in Table IV is unchanged. Extended precision is therefore not a requirement of the one-body half at all: its one-dimensional blocks are evaluated exactly because they are small enough for that to cost nothing, not because a congruence forces it. What does still force it is the  $V_{ee}$  re-basing, which accordingly accounts for essentially the whole cost of the calculation.

[**MEASURED**] What the residual  $\sim 0.4$  mHa is *not* is more azimuthal channels:  $\varphi$  buys under +0.06 pp on top of  $\delta$ . What it *is* remains unattributed, and per the elimination statement above it must not be quoted as the electron–electron cusp: radial completeness and the cusp are the two standing candidates, both converging as the slow  $L^{-3}$  partial-wave crawl, and the knob elimination does not separate them. A literal 99.9% therefore needs a much larger basis or explicit correlation (geminals); what is established is that a *radial* re-basing within this family does not reach it.

[**MEASURED**] The  $\mu$ -weight-adapted family—the generalized Laguerre  $L_n^{(\mu)}$  for the  $(\xi^2 - 1)^{\mu/2}$  weight and the Gegenbauer  $C_n^{(\mu+1/2)}$  for  $(1 - \eta^2)^{\mu/2}$ , reducing to plain Laguerre $\times$ Legendre at  $\mu = 0$ —spans the identical space (the energy agrees to  $< 1$   $\mu$ Ha at every truncation; at  $(5,5) + \delta$  and the optimal  $\alpha = 1.40$  the two families agree to all seven printed digits,  $E = -1.1741513$  Ha) at a normalized condition number *six* orders of magnitude smaller ( $3.7 \times 10^4$  vs  $4.9 \times 10^{10}$  at  $(5,5) + \delta$ ,  $\alpha = 1.40$ ;  $9.1 \times 10^4$  vs  $9.4 \times 10^{10}$  at  $\alpha = 1.0$ ), which is what keeps the downcast solve trustworthy at large truncation. [**MEASURED**] Earlier versions of this sentence read “two to

three orders of magnitude” while quoting the  $(5, 5) + \delta$  pair, whose ratio is  $1.0 \times 10^6$ ; the “two to three” is the  $326\times$  of the  $(3, 3)$ ,  $|m| \leq 1$  case and does not describe the figures it was attached to. It does *not* reach a given accuracy with fewer functions: at equal degree the span, and hence the energy, is the same—the gain is conditioning alone. A per-*function* exponent instead (making the set  $L^2$ -orthonormal) would break completeness and plateau the accuracy, and is not used.

Backing: module `geovac/prolate_recondition.py`, tests `tests/test_paper12_recondition.py`.

## B. Explicit correlation, evaluated algebraically

The knob-elimination above closes with an explicit alternative: a literal 99.9% “needs a much larger basis or explicit correlation (geminals).” The second route is available inside this paper’s own machinery, and it does not leave the algebraic (quadrature-free) discipline of Sec. II. Multiply each product function by a power of the interelectronic distance,  $\Phi = g(\xi_1, \eta_1; \xi_2, \eta_2) r_{12}^p$  with  $p \in \{0, 1\}$  (a James–Coolidge basis in prolate coordinates), and every resulting matrix element—overlap,  $V_{ne}$ ,  $V_{ee}$ , and kinetic—reduces to the *same* algebraic objects already built for  $V_{ee}$ .

[**MEASURED 2026-09-20**] The reduction is exact. Writing  $r_{12}^2 = (R/2)^2(A - B \cos \Delta\varphi)$  with  $A, B$  the standard prolate combinations, every *even* power of  $r_{12}$  is a polynomial in the separated coordinates (its azimuthal average is a finite sum of the one-electron moments of Sec. II), and every *odd* power is  $r_{12}^{2s+1} = (R/2)^{2s+2}(A - B \cos \Delta\varphi)^{s+1}/r_{12}$ , whose azimuthal average is the same  $AK_0 - BK_1$  combination of Neumann kernels—the  $m = 0$  and  $m = 1$  channels of the general- $m$   $V_{ee}$  table of Sec. II—that the Coulomb term itself uses. The one element that looks harder, the kinetic coupling between a  $p = 0$  and a  $p = 1$  function, collapses: integrating  $\frac{1}{2} \langle \nabla g_u \cdot \nabla (g_v r_{12}) \rangle$  by parts cancels the gradient-of- $r_{12}$  term against half of itself and leaves

$$T_{p_0 \times p_1} = -\frac{1}{2} \int r_{12} g_v (\nabla^2 g_u) dV, \quad (33)$$

a pure odd- $r_{12}$  element (no  $\nabla r_{12}$ , no non-analytic term) with  $\nabla^2 g_u$  a polynomial numerator, routed through the same kernels. Each block was validated against direct quadrature to  $\sim 10^{-4}$  (grid-limited).

[**MEASURED 2026-09-20**] On the  $\sigma$  ( $\mu = 0$ ) gerade basis this reaches 99.97% of  $D_e$  (0.053 mHa) at  $(j_{\max}, l_{\max}) = (3, 4)$ ,  $n = 416$ ,  $\alpha = 1.0$ —a sixfold reduction of the residual from the re-based CI’s 0.32 mHa, obtained with exact algebraic integrals (no quadrature at any stage). The energy descends monotonically from above along nested truncations ( $-20.7$ ,  $-0.167$ ,  $-0.079$  mHa at  $n = 18, 90, 160$ ), and the extended precision of Sec. VIIA is again load-bearing: by  $n \simeq 400$  the overlap conditioning reaches  $\sim 10^{19}$  and the double-precision solve discards genuine

directions (16  $\mu$ Ha above the extended-precision energy), while the mpf-orthogonalized solve retains them.

[**MEASURED 2026-09-20**] This constructively settles what the Sec. VIIA residual *is*. It is electron–electron correlation, captured by ordinary integer powers of  $r_{12}$ —not the non-analytic  $r_{12}^{1/2}$ ,  $r_{12} \ln r_{12}$  terms this paper once invoked (Abstract, [**WITHDRAWN**]). The explicit- $r_{12}$  energy then *plateaus* near 0.05 mHa: at  $p \leq 1$  and a single  $\alpha$  the overlap becomes linearly dependent (cond  $\sim 10^{19}$  with essentially all directions retained), i.e. that space is nearly spanned; a variational  $\alpha$  scan gains only  $\sim 2 \mu$ Ha (its optimum is  $\alpha \approx 1.0$ – $1.2$ , and the CI optimum  $\alpha = 1.40$  is *worse* here, because explicit  $r_{12}$  already supplies the correlation a contracted exponent was compensating for). The microhartree regime is therefore not an  $\alpha$  or radial-degree question but a higher- $r_{12}$ -power ( $p \geq 2$ ) one, which the same  $AK_0 - BK_1$  reduction extends to (with additional  $m \geq 2$  channels). We report the  $p \leq 1$  result and its measured ceiling, not a microhartree claim.

[**MEASURED**] The construction is not specific to  $\text{H}_2$ . Because the  $r_{12}$  kernels touch only the electron coordinates, the whole reduction is charge-independent: the sole term that changes for a *heteronuclear* two-electron diatomic is  $V_{ne}$ , whose per-electron form  $(2/R)[(Z_A + Z_B)\xi + (Z_B - Z_A)\eta]/(\xi^2 - \eta^2)$  stays polynomial after the Jacobian (the  $(Z_B - Z_A)\eta$  term breaks the  $\eta \rightarrow -\eta$  symmetry, so both angular parities enter and  $V_{ne}$  couples them). Applied to  $\text{HeH}^+$  at the reference geometry  $R_e = 1.463 a_0$  the same engine converges variationally to 0.9 mHa of the Kolos–Peek reference [15] (well depth  $16456 \text{ cm}^{-1}$ ,  $E_{\text{BO}}(R_e) = -2.9787 \text{ Ha}$ ) at a modest basis, and—consistently with the  $\text{H}_2$  plateau above—the lever there is the *angular* basis, not the exponent (a two-block exponent gains  $\sim 0.3$  mHa, one more angular shell  $\sim 11$  mHa). Scanning  $R$  locates the bond length as well:  $R_{\text{eq}}$  converges from  $-4.5\%$  at  $l_{\max} = 2$  to  $-0.5\%$  at  $l_{\max} = 3$  ( $1.456$  vs  $1.463 a_0$ ), by the *same* angular lever—so the engine reproduces the heteronuclear *geometry*, not only the energy, and the residual is heteronuclear polarization (an angular effect). This is the two-center, two-different-charge analogue of the  $\text{H}_2$  convergence, and the precedent for the frozen-core LiH extension.

[**MEASURED 2026-09-21**] The reduction is also not specific to two electrons, and this places the natural boundary of the quadrature-free, *resolution-of-identity-free* discipline. For a James–Coolidge ansatz  $\Psi = F\Phi$  with  $F = \sum_{p < q} f_{pq}$  multiplicative, the variational kernel  $\langle \Phi | F H F | \Phi \rangle$  stays at most three-body through  $N = 3$ : every term of the operator product closes with closed-form angular rules (a shared-vertex rule fixing scalar legs to  $L = 0$ , and the triangle rule  $\langle P_a(12)P_b(13)P_c(23) \rangle = \delta_{abc}/(2a + 1)^2$ ), so the three- and four-electron integrals that force a resolution of identity in Gaussian  $R_{12}/F_{12}$  theory are here evaluated *exactly*, with no such approximation. At  $N = 4$  a genuinely four-body term first appears—the disjoint-pair Coulomb chain  $f_{12} f_{34} r_{13}^{-1}$ , with the two correlation edges bridged by one Coulomb,

four distinct electrons (only the two-body Coulomb can bridge disjoint pairs; the kinetic gradients are pair-local and  $V_{ne}$  is one-body). This term is nonetheless still *exact*, still RI-free, and still *reducible*: the bridging-Coulomb multipole sum terminates at  $L \leq 2l_{\text{bridge}}$  (measured  $s \rightarrow 0, p \rightarrow 2, d \rightarrow 4$ , bounded by the finite orbital angular content, not by the Coulomb), and the Legendre addition theorem factorizes the chain across the bridge into a finite  $(L, M)$  contraction of vertex kernels—the chain analogue of the triangle contraction. A native evaluation was validated against brute-force 12-dimensional Monte-Carlo integration to  $2 \times 10^{-4}$  on the Be-relevant four-electron integral, including with the antisymmetric determinant (exchange). As a proof of concept it was exercised inside a minimal-basis Be  $R_{12}$ -CI, where a single short-range geminal captures  $\sim 19\%$  of the correlation energy variationally with the four-body term evaluated by this reduction. The residual hard walls are on separate axes—an explicit four-body correlation operator is a four-body Pauli string in a qubit encoding, and the non-Hermitian transcorrelated route carries vector legs that do not reduce this way—but the classical exactness axis, on which Gaussian  $F_{12}$  resorts to a resolution of identity, extends past  $N = 3$  rather than stopping there. This is a diagnostic result at proof-of-concept resolution, not a spectroscopic  $N = 4$  claim.

**[MEASURED 2026-09-21]** Two extensions sharpen this boundary and carry the RI-free discipline from a standalone integral into a genuine many-electron correlation calculation. *First*, the four-body bridge is not tied to the single-center atom: evaluated between two-center prolate densities it reduces by the *same* construction—an isotropic one-dimensional leaf dressing followed by a two-center prolate Neumann Coulomb between the dressed densities—and reproduces an independent twelve-dimensional Monte-Carlo evaluation to  $\sim 2 \times 10^{-5}$  for *every* azimuthal channel ( $\sigma, \pi, \delta$ ; the  $\pi$  channel, which dominates valence correlation, included), with the self-Coulomb step anchored to the closed form  $5\alpha/8$ . *Second*, the same machinery evaluates the many-body matrix elements of an explicit- $r_{12}$  CI on a two-center four-electron reference ( $\Phi_0 = |1s_A^2 1s_B^2|$ , a proof-of-concept ionic reference, not a spectroscopic LiH) with no resolution of identity, each element checked against an independent variational Monte-Carlo ground truth. The entire off-diagonal coupling  $h = h_T + h_{V_{ne}} + h_{V_{ee}}$  reduces to grid integrals of dressed one-electron densities and matches the Monte-Carlo values to within their statistical error: the kinetic part via an integration-by-parts identity that collapses the gradient coupling  $\nabla F \cdot \nabla \Phi_0$  to a *scalar* screened-Coulomb (Yukawa) expectation ( $\nabla^2 f = \gamma^2 f - 2\gamma e^{-\gamma r}/r$ ), the Coulomb part via the two-center Neumann *potential* of the transition density; the diagonal one-body moment  $\langle \chi^2 V_{ne} \rangle$  ( $\chi = F - \bar{F}$ ) reduces by a block decomposition  $F = S + I$  (intra/inter) of the reference density with all three ( $\langle S^2 \rangle$ ,  $\langle SI \rangle$ ,  $\langle I^2 \rangle$ ) parts Monte-Carlo-validated. Crucially, the four-body bridge appears *inside* the diagonal  $V_{ee}$  mo-

ment as  $\langle f_{12} f_{34} r_{13}^{-1} \rangle = \frac{1}{4} I_{\text{coul}}[S^f, S^f]$ —the Coulomb self-interaction of the  $f$ -dressed one-electron slice, via the same Neumann potential—and again matches Monte-Carlo to within statistical error, so the RI-free treatment of the genuinely four-body content is demonstrated not merely as an isolated integral but inside a real correlation quantity. (A technical enabler: the screened-Coulomb dressing fields needed here are obtained without a Yukawa-specific Green’s function, as  $\Psi^Y = \Psi^{\text{coul}} - \Psi^s$  with the non-singular kernel  $s(d) = (1 - e^{-\gamma d})/d$ .) The remaining boundary is now sharper than “ $N = 4$ ”: completing the diagonal  $\langle \chi^2 V_{ee} \rangle$  requires a two-center *three-body triangle*  $\langle f_{13} f_{23} r_{12}^{-1} \rangle$  (two correlation edges sharing a vertex, closed by a Coulomb), whose down-integration yields a *non-separable* two-point kernel  $H_2(1, 2) = \int \rho(3) f(r_{13}) f(r_{23}) d^3 r_3$ ; a low-rank expansion of  $H_2$  followed by a per-mode Neumann Coulomb is the identified next reduction, and the full analytic assembly of the CI energy awaits it. Until then the assembled energy is available from the variational Monte-Carlo route alone, which places the explicit- $r_{12}$  correction correctly signed and above the exact value.

Backing: test `tests/test_paper12_r12.py` and the reduction of Eq. (33) above; the two-center bridge and the many-body matrix-element reductions are validated against variational Monte-Carlo in the sprint tree and chronicled in `CHANGELOG.md` (v5.15.10–v5.15.12). (The prototype engine migrates from the sprint tree into `geovac/`, with dedicated regression tests, when the triangle reduction closes the analytic energy.)

## VIII. IMPLICATIONS FOR THE GEOVAC PROGRAM

### A. The natural geometry hierarchy

Papers 0, 1, and 7 established  $S^3$  as the natural geometry for one-center, one-electron systems. Paper 11 established prolate spheroids as the natural geometry for two-center, one-electron systems. This paper completes the analysis for two-center, two-electron systems and identifies the hierarchy of natural geometries:

1. **One-center, one-electron** (H,  $\text{He}^+$ ): Three-sphere  $S^3$  via Fock projection. The Schrödinger equation becomes a Laplace–Beltrami eigenvalue problem. Accuracy:  $< 0.1\%$  [1, 2].
2. **Two-center, one-electron** ( $\text{H}_2^+$ ): Prolate spheroidal coordinates  $(\xi, \eta, \varphi)$ . The Schrödinger equation separates into two coupled ODEs. Accuracy: machine precision (spectral Laguerre) [4].
3. **Two-center, two-electron** ( $\text{H}_2$ ): Prolate spheroidal  $\times$  prolate spheroidal with Neumann-algebraic  $V_{ee}$ . One-electron physics is exact; electron correlation is resolved to 92.4% in the  $\sigma$  sector, 99.1% once the azimuthal channels

are restored (Sec. VII), and 99.81–99.97% after re-conditioning and the explicit- $r_{12}$  extension (Secs. VII A, VII B). *This paper*.

Each level in this hierarchy addresses a qualitatively different physical singularity: the nuclear Coulomb singularity ( $1/r$ ), the two-center potential ( $1/r_A + 1/r_B$ ), and the electron-electron Coulomb singularity ( $1/r_{12}$ ). The GeoVac program identifies each singularity’s natural coordinate system and discretizes the Laplace–Beltrami operator in that system.

**[WITHDRAWN]** A fourth entry appeared here in earlier versions—“three-body coalescence: hyperspherical coordinates for the interelectron cusp, identified but not yet implemented”—presented as a consequence of this paper’s residual. It is removed, because the residual was misdiagnosed (Sec. VI). This is a statement about what *this paper* licenses, not about hyperspherical coordinates, whose use for two-electron atoms rests on separate grounds in Paper 13 [5]. What the present result removes is the *evidence* that fourth entry rested on, not a verdict in its favour. Comparisons between the two geometries are meaningful only at matched  $m_{\max}$ , and at matched  $m_{\max}$  neither this paper nor Paper 15 [6] claims an advantage over the other. The two published figures are not a matched pair: the 99.1% here is  $(j_{\max}, l_{\max}) = (3, 3)$  at  $|m| \leq 1$ , while the 96.0% there is  $l_{\max} = 6$  with a Schwartz cusp correction and a different solver class, so at matched  $m_{\max}$  neither claims an advantage over the other. Beyond that matched comparison, re-basing the  $|m| \leq 1$  span onto an orthogonal-polynomial family (Sec. VII A) takes this paper’s  $H_2$  to 99.81% of  $D_e$ , and the explicit- $r_{12}$  extension (Sec. VII B) to 99.97% (0.053 mHa)—which settles constructively that the residual is electron correlation captured by ordinary integer powers of  $r_{12}$ , not an inability of these coordinates to represent the cusp. The defensible statement is the negative one: prolate spheroidal coordinates are not disqualified for  $H_2$  by an inability to represent the cusp, because no such inability is in evidence.

## B. Quantum chemistry without six-dimensional quadrature

The Neumann algebraic  $V_{ee}$  realizes a form of quantum chemistry in which the two-electron integrals are reduced entirely to recurrence relations and closed forms on quantum number labels—with the log-singular second-kind moment  $B_l$  now closed-form (Sec. III D), *no* numerical integration remains in either the  $\sigma$  or the  $|m| \geq 1$  sector. Combined with the analytical kinetic energy (integration by parts) and nuclear attraction (simple polynomial moments), the *entire* Hamiltonian matrix is constructed without any quadrature at all.

This has conceptual significance beyond computational efficiency: it demonstrates that the physics of electron-electron repulsion in prolate spheroidal coordinates is

fully encoded in the algebraic structure of the Legendre functions  $P_l$  and  $Q_l$ , not in the spatial geometry of a quadrature grid. The Neumann expansion is not an approximation; it is the exact representation of  $1/r_{12}$  in the prolate spheroidal basis, truncated only by the angular momentum selection rule imposed by the basis itself.

## C. Comparison with the $S^3$ approach

The graph Laplacian FCI approach (Papers 0, 1, 7, and the FCI papers) constructs  $V_{ee}$  from Slater  $F^0$  integrals on the  $S^3$  density overlap [3], which are computed analytically from hydrogen radial wavefunctions. That approach also avoids numerical integration of  $V_{ee}$  and achieves comparable accuracy for atoms.

The prolate spheroidal approach with Neumann  $V_{ee}$  extends this philosophy to molecules: the natural coordinate system for each class of problems determines the algebraic structure of the two-electron integrals. On  $S^3$ , the algebraic structure is the Slater expansion in radial quantum numbers. On the prolate spheroid, it is the Neumann expansion in Legendre orders. Both are special cases of the general principle: *in the natural coordinate system, the Coulomb kernel has a known multipole expansion that converts integrals into algebraic sums.*

## IX. CONCLUSION

We have demonstrated that the Neumann expansion of  $1/r_{12}$  in prolate spheroidal coordinates eliminates the electron-electron integration bottleneck in the two-electron prolate spheroidal CI for  $H_2$ . The key results are:

1. **92.4% of exact  $D_e$**  with fully quadrature-free algebraic  $V_{ee}$  (the log-singular  $B_l$  moment closed-form, Sec. III D) and zero fitted parameters.
2. **12–20 percentage point improvement** over numerical  $V_{ee}$  at every basis size from 1 to 72 functions.
3. **Numerical  $V_{ee}$  saturates at ~80%** regardless of basis size, confirming grid truncation error as the dominant error source.
4. **27 basis functions is the sweet spot:** 92.2% in 705 s versus 92.4% in 5271 s at 72 functions.
5. **The 7.6% gap** is a basis-space limit, not an integration error—and specifically a *configuration* limit: the  $\varphi$ -independent basis spans only  $m_1 = m_2 = 0$ , while a  $^1\Sigma_g^+$  state constrains only  $M = m_1 + m_2$ . The  $\pi^2$  and  $\delta^2$  configurations it excludes carry the angular correlation.
6. **99.1% of exact  $D_e$**  (stability envelope 99.0–99.1%) on restoring the azimuthal channels at

$|m| \leq 1$  in the same basis with the same Neumann kernel (Sec. VII), a gain of 11.6 mHa, against 0.34 mHa from near-tripling the  $\sigma$  basis. The  $\sigma$ -only value reproduces independently in a Gaussian basis to 0.2 mHa, and the extended value to within the stated stability envelope.

7. 99.81% of exact  $D_e$  (0.32 mHa, chemical accuracy) once that 99.1% monomial basis is re-conditioned and the shared exponent is taken to its variational optimum  $\alpha = 1.40$  (99.77% at the fixed  $\alpha = 1.0$  of Table IV): its  $\xi^j$  radial factor is a Hankel moment problem (the 99.1% is a conditioning wall, not a ceiling), and re-basing the same span onto orthogonal polynomials is an exact change of basis that lets the energy climb monotonically and variationally to chemical accuracy at  $(5, 5) + \delta$  (Sec. VII A). What the residual is *not* is more azimuthal channels; radial completeness and the  $e$ - $e$  cusp remain candidates that the knob elimination does not separate.

[**WITHDRAWN**] Earlier versions of this paper concluded that the residual was the electron-electron cusp demanding non-analytic  $r_{12}^{1/2}$  and  $r_{12} \ln r_{12}$  terms, and read that as identifying hyperspherical coordinates as the next natural geometry. Both are withdrawn: the calculations above contain no such term and reach chemical accuracy—0.053 mHa with the explicit- $r_{12}$  extension—and an accurate grid-based prolate spheroidal treatment in the literature [7] independently shows these coordinates are not cusp-limited.

The result demonstrates a form of quantum chemistry where recurrence relations and closed forms on quantum number labels replace the six-dimensional quadrature for electron-electron repulsion entirely—including the log-singular  $B_l$  moment, whose closed form (Sec. III D) carries the same construction to the associated Legendre functions of the  $|m| \geq 1$  channels (Sec. VII), so no numerical integration remains. Just as Paper 7 showed that atomic energies are encoded in the topology of  $S^3$ , this paper shows that molecular electron-electron repulsion is encoded in the algebraic structure of the Legendre functions  $P_l$ ,  $Q_l$  and their associated forms  $P_l^m$ ,  $Q_l^m$  on the prolate spheroid. The physics is in the algebra, not in the grid.

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## Appendix A: Numerical Verification of Auxiliary Integrals

All auxiliary integrals were validated against independent numerical quadrature. The test suite

(`tests/test_neumann_ven.py`, 32 tests) verifies:

- $C_l(q)$ : Base cases ( $C_0$  even/odd,  $C_1$  odd), selection rule, and known analytical values. Agreement to machine precision ( $< 10^{-14}$ ).
- $A_n(\alpha)$ : Comparison with `scipy.integrate.quad`. Agreement to  $< 10^{-12}$ .
- $A_l(p, \alpha)$ : Comparison with numerical quadrature of  $\xi^p e^{-\alpha \xi} P_l(\xi)$ . Agreement to  $< 10^{-10}$ .
- $B_l(p, \alpha)$ : Comparison with independent adaptive quadrature. Agreement to  $< 10^{-10}$ .
- $X_l(p_1, p_2, \alpha)$ : Comparison with 2D numerical quadrature. Agreement to  $< 10^{-8}$ . Symmetry  $X_l(p_1, p_2) = X_l(p_2, p_1)$  verified to  $< 10^{-14}$ .
- $V_{ee}$  **matrix**: Symmetry, positive diagonal, and comparison with numerical elliptic integral quadrature. Neumann vs. numerical agreement to  $< 0.01$  Ha per element (the discrepancy is the grid error in the numerical values).
- **Full Hamiltonian**: Variational principle (energy above exact), bound state, and consistency between Python and Numba code paths ( $< 10^{-8}$  Ha).

## Appendix B: Comparison with $r_{12}$ -Dependent Basis Functions

We tested basis functions with explicit  $r_{12}^p$  dependence ( $p = 1, 2$ ), following the James-Coolidge approach [13]. The  $r_{12}$  factor is expressed in prolate spheroidal coordinates as a function of  $(\xi_1, \eta_1, \xi_2, \eta_2, \varphi_1 - \varphi_2)$ , making the integrals five-dimensional after azimuthal averaging.

Table V compares the best  $r_{12}$ -dependent results with the  $p = 0$  Neumann results.

TABLE V. Comparison of  $r_{12}$ -dependent basis (numerical  $V_{ee}$ ) vs.  $p = 0$  basis (Neumann  $V_{ee}$ ). All results at  $R = 1.4011$  bohr.

| Method                                     | $N$ | $D_e\%$ | $t$ (s) |
|--------------------------------------------|-----|---------|---------|
| $r_{12}^1$ , num., $60 \times 40$ grid     | 6   | 85.7    | 447     |
| $r_{12}^{0-2}$ , num., $20 \times 14$ grid | 18  | 86.8    | 970     |
| $p=0$ , Neumann, $30 \times 20$ grid       | 27  | 92.2    | 705     |
| $p=0$ , Neumann, $30 \times 20$ grid       | 72  | 92.4    | 5271    |

The Neumann  $p = 0$  result at 92.2% exceeds the best  $r_{12}$ -dependent numerical result (86.8%) by 5.4 percentage points, confirming that exact  $V_{ee}$  evaluation is more important than  $r_{12}$ -dependent basis functions when the  $V_{ee}$  integrals are computed numerically. The combination of  $r_{12}$ -dependent basis functions with Neumann-type algebraic  $V_{ee}$  is a natural extension left for future work.

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